measured. And neither Bohm, Einstein, Podolsky, nor Rosen would have argued that the measurements can be done. Instead, they argued that the particles possess features forbidden by the uncertainty principle even though we can never explicitly know their particular values. Such features have come to be known as *hidden features*, or, more commonly, *hidden variables.*

Here is where John Bell changed everything. He discovered that even if you can't actually determine the spin of a particle about more than one axis, still, if in fact it *has* a definite spin about all axes, then there are testable, observable consequences of that spin.

Reality Testing

To grasp the gist of Bell's insight, let's return to Mulder and Scully and imagine that they've each received another package, also containing titanium boxes, but with an important new feature. Instead of having one door, each titanium box has three: one on top, one on the side, and one on the front.[13] The accompanying letter informs them that the sphere inside each box now randomly chooses between flashing red and flashing blue when any one of the box's three doors is opened. If a different door (top versus side versus front) on a given box were opened, the color randomly selected by the sphere might be different, but once one door is opened and the sphere has flashed, there is no way to determine what would have happened had another door been chosen. (In the physics application, this feature captures quantum uncertainty: once you measure one feature you can't determine anything about the others.) Finally, the letter tells them that there is again a mysterious connection, a strange entanglement, between the two sets of titanium boxes: Even though all the spheres *randomly* choose what color to flash when one of their box's three doors is opened, if both Mulder and Scully happen to open the *same* door on a box with the *same* number, the letter predicts that they will see the same color flash. If Mulder opens the top door on his box 1 and sees blue, then the letter predicts that Scully will also see blue if she opens the top door on her box 1; if Mulder opens the side door on his box 2 and sees red, then the letter predicts that Scully will also see red if she opens the side door on her box 2, and so forth. Indeed,